ModelIt!

Teacher's Guide

Can We Run Out of Rare Earth Metals?

Mining, Recycling, and the Hidden Cost of Modern Technology

Quick Reference

Grade Level	Grade 07L2
NGSS Standard	MS-ESS3-1, MS-ESS3-4
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-ESS3-1, MS-ESS3-4

Construct a scientific explanation based on evidence for how the uneven distributions of Earth's mineral, energy, and groundwater resources are the result of past and current geoscience processes. Construct an argument supported by evidence for how increases in human population and per-capita consumption of natural resources impact Earth's systems.

Three-Dimensional Learning

Science Practice	**Constructing Explanations and Designing Solutions** — Students construct scientific explanations for how resource distribution results from geoscience processes and design solutions for sustainable resource management.
Disciplinary Core Idea	**ESS3.A Natural Resources / ESS3.C Human Impacts on Earth Systems** — Humans depend on Earth's mineral resources, which are unevenly distributed. Increases in consumption impact Earth's systems and resource availability.
Crosscutting Concept	**Cause and Effect** — Students trace the causal relationships between mining rate, recycling rate, available supply, market price, and substitution incentive to understand how human resource decisions create cascading effects.

CAST Testing Alignment

MULTIPLE CHOICE:

A country that depends on imported rare earth metals for its technology industry faces a supply disruption. According to the supply-price-substitution model, what would happen first?

CONSTRUCTED RESPONSE:

Using your resource model, explain why simply finding new rare earth deposits is NOT a long-term solution to the supply problem. Describe how a circular economy approach using recycling and substitution provides a more sustainable path, supported by evidence from your simulation scenarios.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Rare earth metals are the hidden ingredients in nearly every piece of modern technology. Despite their name, they are not actually rare, but they are rarely found in concentrations high enough to mine profitably. As global demand skyrockets for electric vehicles, smartphones, and renewable energy technology, the question of whether we can sustain our consumption has become urgent.

What Are Rare Earth Metals?

Rare earth metals are a group of 17 elements including neodymium (magnets in headphones and wind turbines), lanthanum (camera lenses and batteries), and cerium (catalytic converters and glass polishing). They are called 'rare' not because they are scarce in Earth's crust, but because they rarely form concentrated deposits. They tend to be dispersed thinly across rock formations, making extraction difficult and expensive.

Where They Come From

Rare earth deposits formed through geoscience processes over billions of years. Magmatic activity concentrated certain elements in specific rock formations. Weathering and erosion redistributed them. Today, major deposits are found in China (which controls about 60% of global mining), Australia, the United States, Myanmar, and a few other countries. This uneven distribution creates geopolitical vulnerabilities.

The E-Waste Problem

About 50 million tons of electronic waste are generated globally each year, containing billions of dollars worth of rare earth metals. Yet less than 1% of rare earths are recovered from e-waste. The recycling challenge is that rare earths are used in tiny amounts spread across many components, making extraction expensive. A typical smartphone contains about 0.05 grams of rare earths across dozens of different parts.

The Path to Sustainability

Three strategies can address the rare earth challenge: (1) Increase recycling technology to economically recover rare earths from e-waste, (2) Develop substitute materials that can replace rare earths in key applications, and (3) Reduce consumption through more efficient product design and longer product lifespans. A circular economy that recycles, substitutes, and reduces can extend rare earth supplies for centuries.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L — Locate the System
Students identify mining rate, recycling rate, available supply, market price, and substitution incentive as components of the rare earth resource system.

E — Establish Relationships
Students determine that mining rate and recycling rate are external inputs that drive available supply, which cascades to affect market price and substitution incentive.

V — Visualize & Model
Students build a model showing how human resource decisions (mining and recycling) determine the sustainability of rare earth metal supply.

E — Evaluate Outcomes
Students run business-as-usual, circular economy, and supply crisis scenarios to observe how supply, price, and substitution incentive respond to different strategies.

R — Revise & Extend
Students add geopolitical risk or technological demand growth to explore how external pressures complicate resource management.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Pick up your phone. Inside it are metals that were mined from deep underground, possibly on the other side of the world. What happens when those mines run dry?"*

Watch For

Pass around a disassembled circuit board. Ask: How many different materials do you think are in this?

Slide 2: Learning Objectives

3 min

Say: *"Today we're investigating whether the technology we love is sustainable, or if we're heading toward a resource crisis."*

Watch For

Have students read objectives. Pre-teach rare earth metals and circular economy concepts.

Slide 3: Big Question

3 min

Say: *"Every electric car battery, every wind turbine, every MRI machine uses rare earth metals. Demand is skyrocketing. Supply is limited. What happens next?"*

Watch For

Show infographic of rare earth metals in everyday devices. Discuss which devices students use daily.

Slide 4: LEVER Framework

2 min

Say: *"We'll build a model of Earth's rare earth metal supply chain."*

Watch For

Preview LEVER steps. Show world map of rare earth deposits.

Slide 5: Activity 1: Components

8 min

Say: *"What factors determine whether we'll have enough rare earth metals? Which can we control?"*

Watch For

Guide sorting. Discuss why mining rate and recycling rate are the human-controlled inputs.

Slide 6: Activity 2: Connections

8 min

Say: *"When mining goes up, what happens to supply? When supply goes down, what happens to prices? And when prices go up, what happens to the search for alternatives?"*

Watch For

Help students trace the economic feedback loop. Emphasize cause-and-effect chains.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's mine like there's no tomorrow and see what happens. Then let's try recycling and see the difference."*

Watch For

Students predict first, then run all three scenarios. Compare long-term supply projections.

Slide 8: Discoveries

5 min

Say: *"So will we actually run out? What did our model reveal about the real solution?"*

Watch For

Lead discussion about circular economy vs. linear consumption. Connect to Earth's finite resources.

Slide 9: STEM Challenge

12 min

Say: *"Your city needs a recycling program that recovers these valuable metals. Design one using your data."*

Watch For

Groups design community recycling programs with collection, extraction, and impact projections.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Mining Rate, Recycling Rate
- **INTERNAL:** Available Supply, Market Price, Substitution Incentive

Relationships

- **Mining Rate to Available Supply:** POSITIVE (+) — Higher mining rates extract more rare earth metals from the ground, increasing the available supply for manufacturing.
- **Recycling Rate to Available Supply:** POSITIVE (+) — Higher recycling rates recover rare earth metals from discarded electronics, returning them to the available supply without new mining.
- **Available Supply to Market Price:** NEGATIVE (-) — When available supply increases, market price decreases because supply exceeds demand. When supply drops, scarcity drives prices up.
- **Market Price to Substitution Incentive:** POSITIVE (+) — Higher market prices for rare earth metals create stronger economic incentive to develop cheaper alternative materials that can serve the same function.
- **Mining Rate to Available Supply (long-term):** NEGATIVE (-) — In the long term, high mining rates deplete underground deposits faster, eventually reducing the available supply as easy-to-mine sources are exhausted.

Key Discoveries

- Rare earth metals are not actually rare in Earth's crust, but they are rarely found in concentrations high enough to mine economically
- Currently, less than 1% of rare earth metals are recycled from electronic waste, meaning massive amounts of valuable material end up in landfills
- When Available Supply drops and Market Price rises, it creates economic incentive to develop substitutes, which can reduce dependence on rare earths
- China controls about 60% of rare earth mining and 90% of processing, making global supply vulnerable to geopolitical disruptions

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Rare earth metals are actually rare"	Rare earth metals are NOT rare in Earth's crust. Several are more common than lead or tin. They're called 'rare' because they rarely form concentrated deposits that can be mined economically. The real challenge is extraction, not existence. Earth has plenty of rare earth metals, but most are too spread out to mine profitably.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"We can just find more mines when current ones run out"	Finding new deposits doesn't solve the fundamental problem. Mining creates environmental damage (toxic waste, water contamination, radioactive byproducts) and deposits take millions of years to form. A sustainable solution requires recycling existing materials, not endlessly opening new mines.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Recycling electronics is easy"	Recycling rare earths from electronics is extremely difficult because the metals are used in tiny amounts spread across many different components. A smartphone might contain 0.05 grams of rare earths in dozens of different places. Separating and recovering these materials requires specialized chemical processes that are expensive and technically challenging.